WORKED SOLUTION

STEP I 1987, Question 15

Probability and Statistics • Continuous Random Variables

STEP I 1987, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q15

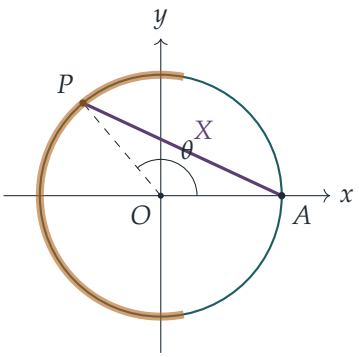
The key idea

"Uniform on the circle" means the *angle* that P makes at the centre is uniformly distributed, so the natural move is to write $P = (\cos\theta, \sin\theta)$ with θ uniform on $[0, 2\pi)$. The distance from $(1, 0)$ is then a chord of the unit circle, and the half-angle formula turns it into a single sine. After that, the mean is one integral, the variance needs almost no integration at all, and the probability is a question about an interval of angles.

Solution

Setting up

Let θ be the angle from the positive x-axis to OP, where O is the origin. Choosing P uniformly on the circle means choosing arc length uniformly, so θ is uniform on $[0, 2\pi)$ with density $\frac{1}{2\pi}$. Write $A = (1, 0)$ and $P = (\cos\theta, \sin\theta)$.



Then

$$X^2 = (\cos\theta - 1)^2 + \sin^2\theta = 2 - 2\cos\theta = 4\sin^2\tfrac{\theta}{2},$$

using $1 - \cos\theta = 2\sin^2\frac{\theta}{2}$. Since $0 \le \frac{\theta}{2} < \pi$, we have $\sin\frac{\theta}{2} \ge 0$, so taking the positive

square root is safe:

$$X = 2\sin\tfrac{\theta}{2}.$$

The mean

$$\mathrm{E}(X) = \frac{1}{2\pi}\int_0^{2\pi} 2\sin\tfrac{\theta}{2}\,\mathrm{d}\theta = \frac{1}{2\pi}\Big[-4\cos\tfrac{\theta}{2}\Big]_0^{2\pi} = \frac{1}{2\pi}(4+4) = \frac{4}{\pi}.$$

So $\mathrm{E}(X) = \dfrac{4}{\pi} \approx 1.273$.

The variance

Ask yourself Do we need to integrate $4\sin^2\frac{\theta}{2}$ to find $\mathrm{E}(X^2)$?

No: the unsquared form is easier. $X^2 = 2 - 2\cos\theta$, and $\cos\theta$ averages to zero over a full turn, since $\int_0^{2\pi}\cos\theta\,\mathrm{d}\theta = 0$. Hence $\mathrm{E}(X^2) = 2$, and

$$\mathrm{Var}(X) = \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = 2 - \frac{16}{\pi^2} \approx 0.379.$$

The probability that X exceeds its mean

We need $2\sin\frac{\theta}{2} > \frac{4}{\pi}$, that is $\sin\frac{\theta}{2} > \frac{2}{\pi}$. Note $0 < \frac{2}{\pi} < 1$, so this has solutions. Let $\alpha = \arcsin\frac{2}{\pi}$, which lies in $(0, \frac{\pi}{2})$. As $\frac{\theta}{2}$ runs over $[0, \pi)$, the sine rises from 0 to 1 and falls back, and it exceeds $\sin\alpha$ exactly when

$$\alpha < \tfrac{\theta}{2} < \pi - \alpha, \qquad \text{that is} \qquad 2\alpha < \theta < 2\pi - 2\alpha.$$

This is the thickened arc in the diagram: the points on the far side of the circle from A. Since θ is uniform, the probability is the length of this interval divided by 2π:

$$\mathrm{P}(X > \mathrm{E}(X)) = \frac{2\pi - 4\alpha}{2\pi} = \boxed{1 - \frac{2}{\pi}\arcsin\frac{2}{\pi}} \approx 0.561.$$

Checks

- X ranges over $[0, 2]$, and the mean $4/\pi \approx 1.27$ sits above the midpoint 1. That fits the picture: the chord length is near its maximum for a large share of the circle (as P passes the far side, X changes slowly), so the distribution is weighted towards large values.
- For the same reason the probability of exceeding the mean is a little more than $\frac{1}{2}$, and 0.561 is.
- Numerically, a midpoint-rule average over 2×10^5 equally spaced angles gives mean 1.273240, variance 0.378861 and $\mathrm{P} = 0.56066$, and a 10^6-point simulation agrees to within sampling error.

Common slips

- Taking $X = 2\sin\frac{\theta}{2}$ with $\theta \in (-\pi, \pi]$. Then $\sin\frac{\theta}{2}$ can be negative, and you need $X = 2\left|\sin\frac{\theta}{2}\right|$; using $[0, 2\pi)$ avoids the modulus.
- Treating the x-coordinate of P as uniform on $[-1, 1]$. Uniform on the circle means uniform in angle (arc length), not in x.

Beyond STEP

This is a first example of transforming a random variable: the density of X is $f(x) = \dfrac{2}{\pi\sqrt{4 - x^2}}$ on $[0, 2]$, an arcsine-type law, and you can recover every answer above from it. The question also hints at how slippery "at random" can be: picking a random chord of a circle in different natural ways gives different answers (Bertrand's paradox). Both ideas are developed in Blitzstein and Hwang, *Introduction to Probability*.

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